

MATERIAL SAFETY DATA SHEET

1. PRODUCT AND MANUFACTURER IDENTIFICATION

BIFENTHRIN 10% W/V

SUPPLIER NAME: Shandong Huimin Vanda Biological Technology Co., Ltd.

SUPPLIER ADDRESS: Jinan Plant Protection Technology Industrial Park, Hanmiao Town, Shanghe County, Jinan City, Shandong Province, China

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CHEMICAL NAME: 2-methylbiphenyl-3-ylmethyl (1RS,3RS)-3-[(Z)-2-chloro-3,3,3-trifluoroprop-1-enyl]-2,2-dimethylcyclopropanecarboxylate
 or
 2-methylbiphenyl-3-ylmethyl (1RS)-cis-3-[(Z)-2-chloro-3,3,3-trifluoroprop-1-enyl]-2,2-dimethylcyclopropanecarboxylate

CAS NUMBER: 82657-04-3

MOLECULAR FORMULA: $C_{23}H_{22}ClF_3O_2$

MOLECULAR WEIGHT: 422,88 g/mol

2.COMPOSITION / INFORMATION ON INGREDIENTS

Bifenthrin 10%

Other ingredients up to 100%

3. HAZARDS IDENTIFICATION

CLASSIFICATION OF THE SUBSTANCE OR MIXTURE

Acute toxicity - Oral, Category 2

Skin sensitization, Category 1B

Acute toxicity - Inhalation, Category 3

Carcinogenicity, Category 2

Specific target organ toxicity – repeated exposure, Category 1

Hazardous to the aquatic environment, short-term (Acute) - Category Acute 1

Hazardous to the aquatic environment, long-term (Chronic) - Category Chronic 1



PICTOGRAMS

Signal Words Danger

Hazard statements

H300 Fatal if swallowed

H317 May cause an allergic skin reaction

H331 Toxic if inhaled

H351 Suspected of causing cancer

H372 Causes damage to organs through prolonged or repeated exposure

H410 Very toxic to aquatic life with long lasting effects

Precautionary statement(s)

P264 Wash ... thoroughly after handling.

P270 Do not eat, drink or smoke when using this product.

P261 Avoid breathing dust/fume/gas/mist/vapours/spray.

P272 Contaminated work clothing should not be allowed out of the workplace.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P271 Use only outdoors or in a well-ventilated area.

P201 Obtain special instructions before use.

P202 Do not handle until all safety precautions have been read and understood.

P260 Do not breathe dust/fume/gas/mist/vapours/spray.

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P273 Avoid release to the environment.

P301+P310 IF SWALLOWED: Immediately call a POISON CENTER/doctor/...

P321 Specific treatment (see ... on this label).

P330 Rinse mouth.

P302+P352 IF ON SKIN: Wash with plenty of water/...

P333+P313 If skin irritation or rash occurs: Get medical advice/attention.

P362+P364 Take off contaminated clothing and wash it before reuse.

P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P311 Call a POISON CENTER/doctor/...

P308+P313 IF exposed or concerned: Get medical advice/ attention.

P314 Get medical advice/attention if you feel unwell.

P391 Collect spillage.

P405 Store locked up.

P403+P233 Store in a well-ventilated place. Keep container tightly closed.

P501: Dispose of contents/container in accordance with local regulations.

4. FIRST AID MEASURES

DESCRIPTION OF FIRST AID MEASURES

GENERAL ADVICE: Consult a physician. Show this safety data sheet to the doctor in attendance.

INHALATION: If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.

SKIN CONTACT: Wash off with soap and plenty of water. Consult a physician.

EYE CONTACT: Flush eyes with water as a precaution.

INGESTION: Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

MOST IMPORTANT SYMPTOMS AND EFFECTS, BOTH ACUTE AND DELAYED: The most important known symptoms and effects are described in the labelling

INDICATION OF ANY IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT NEEDED: No data available

5. FIRE FIGHTING MEASURES

EXTINGUISHING MEDIA: Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

SPECIAL HAZARDS ARISING FROM THE SUBSTANCE OR MIXTURE: No data available

ADVICE FOR FIREFIGHTERS: Wear self-contained breathing apparatus for firefighting if necessary.

6. ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT AND EMERGENCY PROCEDURES

Use personal protective equipment. Avoid dust formation. Avoid breathing vapours, mist or gas. Ensure adequate ventilation. Evacuate personnel to safe areas. Avoid breathing dust.

ENVIRONMENTAL PRECAUTIONS

Prevent further leakage or spillage if safe to do so. Do not let product enter drains. Discharge into the environment must be avoided.

METHODS AND MATERIALS FOR CONTAINMENT AND CLEANING UP

Pick up and arrange disposal without creating dust. Sweep up and shovel. Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

PRECAUTIONS FOR SAFE HANDLING

Avoid contact with skin and eyes. Avoid formation of dust and aerosols.

Provide appropriate exhaust ventilation at places where dust is formed.

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CONDITIONS FOR SAFE STORAGE, INCLUDING ANY INCOMPATIBILITIES

Store in cool place. Keep container tightly closed in a dry and well-ventilated place.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

EYE PROTECTION:	Safety glasses with side-shields. Use equipment for eye protection tested and approved under appropriate government standards
RESPIRATORY PROTECTION:	Where risk assessment shows air-purifying respirators are appropriate use respirator cartridges as a backup to engineering controls. If the full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards
SKIN PROTECTION	Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws. Wash and dry hands.
BODY PROTECTION	Impervious clothing. The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.
CONTROL OF ENVIRONMENTAL EXPOSURE	Prevent further leakage or spillage if safe to do so. Do not let product enter drains. Discharge into the environment must be avoided.
ENGINEERING CONTROLS	Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE FORM:	Transparent liquid
ODOUR	No data available
ODOUR THRESHOLD	No data available
PH	5.32
MELTING POINT/FREEZING POINT	No data available
INITIAL BOILING POINT AND BOILING RANGE	Decomposes before boiling
FLASH POINT	> 70°C
EVAPORATION RATE	No data available
FLAMMABILITY (SOLID, GAS)	No data available
UPPER EXPLOSION LIMIT:	No data available
LOWER EXPLOSION LIMIT:	No data available
VAPOUR PRESSURE	0.0178 mPa
VAPOUR DENSITY	No data available
RELATIVE DENSITY	0.94
WATER SOLUBILITY	Soluble
PARTITION COEFFICIENT: NOCTANOL/WATER	Log P 6.6
SURFACE TENSION	No data available
HENRY'S LAW CONSTANT	No data available
AUTO-IGNITION TEMPERATURE	No highly flammable
DECOMPOSITION TEMPERATURE	280 °C
VISCOSITY	No data available
EXPLOSIVE PROPERTIES	No explosive properties
OXIDIZING PROPERTIES	No oxidizing properties

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY:	Stable under recommended storage conditions.
STORAGE STABILITY:	Stable under normal storage conditions in original container for at least two years.
COMPATIBILITY:	Strong oxidizing agents

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CONDITIONS TO AVOID	No data available
HAZARDOUS DECOMPOSITION PRODUCTS:	Other decomposition products - No data available

11. TOXICOLOGICAL INFORMATION

ACUTE ORAL LD50	LD50 Oral - Rat 5-50 mg/kg bw
ACUTE DERMAL LD50	LD50 Dermal - Rat 1000 mg/kg bw
INHALATION LC50	LC50 Inhalation- Rat 0.52 mg/m ³ air
SKIN IRRITATION	Non-irritant
EYES IRRITATION	Non-irritant
SKIN SENSITISATION	Skin sensitiser (pilot)
SHORT TERM TOXICITY	Relevant oral NOAEL : NOAEL: 2.5 mg/kg/day (90-day dog) NOAEL: 1.5 mg/kg/day (1-year dog) Relevant dermal NOAEL: NOAEL: 50 mg/kg/day (rat) NOAEL: 100 mg/kg/day (rabbits)
DEVELOPMENTAL	> 2 mg/kg bw/d
GENOTOXICITY	No genotoxic potential
REPRODUCTIVE/DEVELOPMENTAL EFFECTS:	5 mg/kg bw/d
CHRONIC/SUBCHRONIC TOXICITY STUDIES:	2-yr rat: NOAEL: 4.7 mg/kg bw/d for males and 3 mg/kg bw/d for females 18-m mice: NOAEL: males: 7.6 mg/kg bw/d & females: 37 mg/kg bw/d
CARCINOGENICITY:	Bladder tumors in male mice (statistically significant at 92 mg/kg bw/d)

12. ECOLOGICAL INFORMATION

TOXICITY

Acute toxicity to birds:
LD50 1800 mg/kg bw/day (Bobwhite quail)
Short-term toxicity to birds:
LD50 = 569 per day (Bobwhite quail)
Acute toxicity fish: Oncorhynchus mykiss
LC50 0.256 mg a.s./L
Long term toxicity fish: S. gairdneri
NOEC 0.012 mg a.s./L
Acute toxicity Daphnia magna,:
EC50: <0.1 mg/kg

PERSISTENCE AND DEGRADABILITY

No data available

BIOACCUMULATIVE POTENTIAL

No data available

13. DISPOSAL CONSIDERATIONS

WASTE TREATMENT METHODS

Product

Offer surplus and non-recyclable solutions to a licensed disposal company. Dissolve or mix the material with a combustible solvent and burn in a chemical incinerator equipped with an afterburner and scrubber.

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CONTAMINATED PACKAGING

Dispose of as unused product.

14. TRANSPORT INFORMATION

IATA

UN-Number 3082

Class 9

PACKING GROUP III

DESCRIPTION OF THE GOODS TOXIC SOLID, ORGANIC, N.O.S. (Bifenthrin)

MARINE POLLUTANT: yes

15. REGULATORY INFORMATION

SAFETY, HEALTH AND ENVIRONMENTAL REGULATIONS/LEGISLATION SPECIFIC FOR THE SUBSTANCE OR MIXTURE

This safety datasheet complies with the requirements of Regulation (EC) No. 1907/2006.

16. OTHER INFORMATION

The data given here is based on current knowledge and experience. The purpose of this safety data sheet is to describe the products in terms of their safety requirements. The data does not signify any warranty with regard to the products' properties.